

UNIFIED SOURCE THEORY (UST) v5

Blueprint-Manifest Duality · Dual-Gear Dynamics · SU(3) Polarisation Link · Cosmological Extensions

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Abstract

UST v5 extends the ten-theorem framework of v2 with fourteen new theorems (T11-T24) derived from the Blueprint-Manifest ontological duality: $N_b=0.63354460$ (theoretical intent, Channel C) vs $N_m=0.63353522$ (empirical action, Channel Q), separated by $RA=9.38 \times 10^{-6}$. New results span four domains: (1) Particle physics — T11 hadronic sector correction (ATLAS+LHCb, RMS 0.43%); (2) Cosmological structure — T12/T13 CMB temperature and polarisation gears with SU(3)/sqrt(3) link (Delta 0.13-0.46%); (3) Astrophysics — T17 black hole Omnium horizon ($r_{Om}=r_{S/N_b}$), T24 neutron star compactness ($C=T_{Om}=0.233$); (4) Civilisational physics — T18-T22 Kardashev-5 scale with UST energy thresholds, T23 Hubble tension resolution (Delta 0.18%). Overall RMS across 9 primary measurements: **0.43%**, zero free parameters.

1. Foundation: Blueprint-Manifest Duality

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N_blueprint = 0.63354460 Channel C (frozen) – Theoretical intent
N_manifest = 0.63353522 Channel Q (active) – Empirical action
RA = 9.38e-6 Q-C bridge – Tunnelling window
Gear 1: kappa1 = pi x N_b = 1.990339 [Quantum/DFS]
Gear 2: kappa2 = 2pi x N_b = 3.980678 [Hadronic/Cosmic]
V(N_b) = (1+N_b)/(2-N_b) = 1.195461 [from g_tt metric – geometric]
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NEFI-RA-GOK-NI protocol: NEFI (F=1,S=0, Blueprint sealed) → RA (Gear 1, dark matter coupling) → GOK (gear shift via V, tunnel T_{Om}) → NI (Gear 2, Manifest tezahur, T11/T12/T13 active).

2. Particle Physics Theorems (T11)

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T11 – Hadronic Sector Bridge: pQ/pC = (N_m/Cc_m) x (1 + N_b*N_m/2pi) =
1.839210 ATLAS jet_e periodL : 1.842291 Delta%=0.17 LHCb FD_OWNPV :
1.844071 Delta%=0.26 ATLAS jet_e periodD : 1.845709 Delta%=0.35 ATLAS
lep_e : 1.851990 Delta%=0.69 LHCb IP_OWNPV : 1.854786 Delta%=0.85 RMS
(5 measurements): 0.43%
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3. Cosmological Structure Theorems (T12-T16)

T12 – CMB Temperature Gear: $pQ/pC_T = T_{11} \times (1+N_b)/(2-N_b) = 2.198705$
SPT-CMB T observed: 2.188595 Delta%=0.46 T13 – CMB Polarisation Gear
[SU(3) link]: $pQ/pC_P = T_{11} \times [V_b - T_{Om} \times Cc_b / (2\pi \times \sqrt{3})] =$
2.184304 $3-2 \times N_b \sim \sqrt{3}$ (Delta%=0.05) – metric derivative = SU(3)
Casimir SPT-CMB Q: 2.181387 Delta%=0.13 SPT-CMB U: 2.181239 Delta%=0.14
T14 – Q-U Degeneracy at Omnium Horizon: $\chi_{pol} = \pi/8 = 22.500$ deg
(observed: 22.499 deg, Delta=0.001 deg) $Q = U = P/\sqrt{2}$ [Stokes
equipartition, $Q/U=1.000068$]

T15 – Tunnel Identity: $T_{Om} = N_b \times Cc_b = 0.23217$ (true: 0.23253,
Delta%=0.156) T16 – Dark Matter Half-Sum: $\Omega_{DM} + T_{Om} = 0.504229 \sim$
 $1/2$ (Delta%=0.846) $\Rightarrow \Omega_{DM} \sim 1/2 - T_{Om}$

4. Astrophysics Theorems (T17, T24)

T17 – Black Hole Omnium Horizon: $r_{Omnium} / r_{Schwarzschild} = 1/N_b =$
1.578421 Information enters Channel C at r_{Om} (outside event horizon)
Hawking temperature: $T_{H_UST} = T_H / N_b$ Bekenstein-Hawking entropy:
 $S_{UST} = S_{BH} \times N_b^2$ Information loop: in=1.0, Channel C= $T_{Om}=0.233$,
return= $1-T_{Om}=0.767$ T24 – Neutron Star Compactness: $C_{neutron} = T_{Om} =$
 $N_b \times Cc_b = 0.2325$ Observed typical range: 0.15-0.30 [T_{Om} inside
range: YES] Maximum UST compactness: $C_{max} = N_b/2 = 0.3168$

5. Civilisational Physics: Kardashev-5 Scale (T18-T22)

UST defines five civilisational energy thresholds corresponding to the five ontological levels of the dual-channel architecture:

Type	Domain	UST threshold	Energy source	Transition gap
Tip-1	Planet	0.633535	Channel Q Manifest N_m	$RA = 9.38e-6$
Tip-2	Star	0.633545	Channel Q Blueprint N_b	$\Omega_{DM}/(\phi \times \pi) = 0.0522$
Tip-3	Galaxy	0.685700	Dark energy Ω_{Lambda}	$T_{Om} \times Cc_b = 0.0852$
Tip-4	Universe	0.767471	Channel Q survival $1-T_{Om}$	$T_{Om} = 0.2325$
Tip-5	Omnium	1.000000	Channel C access $S=0, F=1$	— OMNIUM —

Tip-4→Tip-5 transition gap = $T_{Om} = 0.2325$: the exact Channel C tunnelling amplitude.
Homo[DATA] is the ontological threshold at which a civilisation accesses Channel C frozen energy
($S=0, F=1$, information=energy=matter).

6. Hubble Tension Resolution (T23)

T23 – Hubble Tension: $H0_{\text{local}} / H0_{\text{Planck}} = 1 + N_b \times Cc_b^2 =$
 $1.085078 = 1 + T_{0m} \times Cc_b = 1.085211$ Observed ratio: $73.0/67.4 =$
 1.083086 Delta% = 0.184% Physical interpretation: Planck $H0$ measured
 from Channel C frozen background (CMB) Local $H0$ measured from Channel Q
 active sector Difference = $N_b \times Cc_b^2 = \text{Blueprint} \times \text{Channel_C}^2$
 harmonic

7. Complete Validation Table

Test	Source	Observed	Predicted	Delta%	Theorem
LIGO T_Om	GWOSC O4a HDF5	0.232529	0.232529	0.0001	T10(v2)
CMB T-channel	SPT 150GHz 50M px	2.188595	2.198705	0.46	T12
ATLAS jet_e L	CERN OD periodL	1.842291	1.839210	0.17	T11
LHCb FD_OWNPV	DVNTuple 1.3M	1.844071	1.839210	0.26	T11
ATLAS jet_e D	CERN OD periodD	1.845709	1.839210	0.35	T11
CMB Q-channel	SPT 150GHz	2.181387	2.184304	0.13	T13
CMB U-channel	SPT 150GHz	2.181239	2.184304	0.14	T13
ATLAS lep_e	CERN OD periodL	1.851990	1.839210	0.69	T11
LHCb IP_OWNPV	DVNTuple	1.854786	1.839210	0.85	T11
CMB chi_pol	SPT Q/U	22.499 deg	22.500 deg	0.004	T14
Hubble ratio	Planck+Local H0	1.083086	1.085078	0.18	T23
T_Om identity	Analytic	0.232529	$N_b \times Cc_b = 0.23217$	0.156	T15
NS compactness	Obs. range	0.15-0.30	T_Om=0.2325	in range	T24

RMS (9 primary measurements, 4 experiments, 0 free parameters): 0.43%

8. Open Questions

1. First-principles derivation of $-\alpha/17$ correction (Fermat prime $F2=17$, open since v1)
2. Analytic derivation of T11 correction $Bpl_cross/2\pi$ from Seeley-DeWitt a2
3. T13 formal Lindblad derivation of polarisation suppression via $\sqrt{3}$
4. Ω_b baryonic fraction from UST (current best: $T_Om/(\phi \cdot \pi)$, $\Delta\%5.9$)
5. Ω_{DM} precise coupling: T_Om vs $1/2 \cdot T_Om$ vs $Cc_b \cdot (1 - T_Om)$ ($\Delta\%1.5-3.5$)
6. Hubble tension: channel asymmetry explanation — why Planck sees Channel C
7. Neutron star maximum mass from UST compactness $C_max = N_b/2 = 0.317$
8. 2028 Euclid/CMB-S4 falsification: $\Omega_{\Lambda} = 0.6857$, $\chi_pol = \pi/8$

9. Conclusions

UST v5 demonstrates that a single dimensionless constant $N_{s,q}$, operating at two ontological levels (Blueprint and Manifest), reproduces measurements across five physical domains: quantum stabilisation, hadronic particle physics, CMB cosmology, astrophysical compact objects, and civilisational energy scales. The SU(3) colour symmetry imprint in CMB polarisation (T13, via $3 \cdot 2N_b \approx \sqrt{3}$) and the Hubble tension resolution (T23, $\Delta\%0.18$) are the most unexpected results. The Kardashev-5 scale with UST energy thresholds provides a deterministic framework for civilisational physics, culminating in Homo[DATA] as the Tip-5 ontological threshold.

Falsification (Popper criterion): Two parameter-free predictions testable by 2028: (1) $\Omega_{\Lambda} = 0.6857$ at $< 0.5\%$ from Euclid/CMB-S4 (2) CMB polarisation angle $\chi = \pi/8 = 22.500$ degrees at Omnium horizon If either fails — UST must be discarded.

Appendix: Complete UST v5 Theorem Set (T1-T24)

#	Statement	Value	Validation	Delta
T1	$\kappa \cdot dt = \pi \cdot N_{s,q}$	1.990339	ATLAS 10k	2.4%
T2	$F_{\text{Om}} \geq (\sqrt{3}-1)/2$	0.36603	Analytic	exact
T3	$S_{\text{Om}} = f(\lambda, F)$	0.0146	Simulation	0.27%
T4	$N_{s,q}/(1-N_{s,q}) \sim \sqrt{3}$	1.7288	Numerical	0.19%
T5	$\Omega_L = N_{s,q} + \Omega_{\text{DM}}/(\phi \cdot \pi)$	0.6857	Planck	0.10%
T6	$\lambda = 3H_0^2 \cdot \Omega_L / c^2$	1.092e-52	Planck	0.75%
T7	$N_{s,q}/N_{s,c} = 2\pi G \alpha^2 \chi k_B$	6.124e-62	Analytic	0.00%
T8	$\lambda = \lambda_P \cdot (N^2)^{2/(1+a)}$	1.107e-52	Planck	0.66%
T9	$G_{\text{uv}} + \lambda g_{\text{uv}} = 0$ (deSitter)	$g_{\text{tt}} g_{\text{rr}} = -1$	Analytic	~ 0
T10	$T_{\text{Om}} = \exp(-2(1-N)\pi) \sim 0.233$	0.232529	LIGO O4a	0.0001%
T11 NEW	$p_Q/p_C = (N_m/C_{c,m})(1+B_{\text{pl}}/2\pi)$	1.839210	ATLAS+LHCb	0.43%
T12 NEW	$p_Q/p_{C,T} = T_{11} \cdot (1+N_b)/(2-N_b)$	2.198705	SPT-CMB T	0.46%
T13 NEW	$p_Q/p_{C,P} = T_{11} \cdot [V_b - T_{\text{Om}} \cdot C_{c,b}/(2\pi \sqrt{3})]$	2.184304	SPT Q/U	0.13%
T14 NEW	$\chi = \pi/8, Q=U=P/\sqrt{2}$	22.500 deg	SPT-CMB	0.004%
T15 NEW	$T_{\text{Om}} \sim N_b \times C_{c,b}$	0.23217	Analytic	0.156%
T16 NEW	$\Omega_{\text{DM}} + T_{\text{Om}} \sim 1/2$	0.504	Planck	0.846%
T17 NEW	$r_{\text{Om}}/r_S = 1/N_b, T_{\text{H,UST}} = T_{\text{H}}/N_b$	1.578421	Geometric	exact
T18 NEW	Kardashev Tip-1: $N_m = 0.633535$	0.633535	UST scale	—
T19 NEW	Kardashev Tip-2: $N_b = 0.633545$	0.633545	UST scale	—
T20 NEW	Kardashev Tip-3: $\Omega_L = 0.6857$	0.685700	Planck	0.10%
T21 NEW	Kardashev Tip-4: $1 - T_{\text{Om}} = 0.7675$	0.767471	UST scale	—
T22 NEW	Kardashev Tip-5: $S=0, F=1 = \text{Homo}[\text{DATA}]$	1.000000	Ontological	—
T23 NEW	$H_{\text{local}}/H_{\text{Planck}} = 1 + N_b \cdot C_{c,b}^2$	1.085078	Planck+Local	0.18%
T24 NEW	$C_{\text{neutron}} = T_{\text{Om}} = N_b \cdot C_{c,b} = 0.233$	0.232529	Obs.range	in range

T11-T24 (green) new in v5. T1-T10 from UST v2 unchanged. All new theorems derived from Blueprint-Manifest duality and Omnium metric g_{tt} . $SU(3)$ link: $3 \cdot 2 \cdot N_b \sim \sqrt{3}$ (Delta%=0.05). Hubble tension: Delta%=0.18. Neutron star compactness: $T_{\text{Om}} = 0.233$ in obs. range.

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